WELDING PROCESSES

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Grouping of welding processes

- The grouping of welding processes has been made according to the mode of energy transfer as a primary consideration.

- During the classification, the designation of pressure or nonpressure has been omitted since the factor of pressure is an element of operation of the applicable process.

- Other terms and factors, such as the type of current, whether the electrodes are continuous or incremental or the method of application are not considered.

- **Coalescence:** is defined as the growing together or growth into one body of the materials being welded and is applicable to all types of welding.

Classification of Welding Processes

- Welding Processes
  - Arc welding
    - Nonconsumable Electrode
      - Gas Tungsten
      - Plasma
      - Carbon
      - Stud
    - Consumable Electrode
      - Shielded Metal
      - Gas metal
      - Flux cored
      - Submerged
      - Electroslag
      - Electrogas
  - Oxyfuel welding
    - Oxyacetylene
    - Oxyhydrogen
  - Resistance welding
  - Solid state welding
    - Forge
    - Cold
    - Diffusion
    - Explosion
    - Friction
    - Hot pressure
    - Roll
    - Ultrasonic
    - Coextrusion
  - Others
    - Electron beam
    - Laser beam
    - Thermite
    - Induction
    - Percussion
    - Electroslag

Definition of welding groups

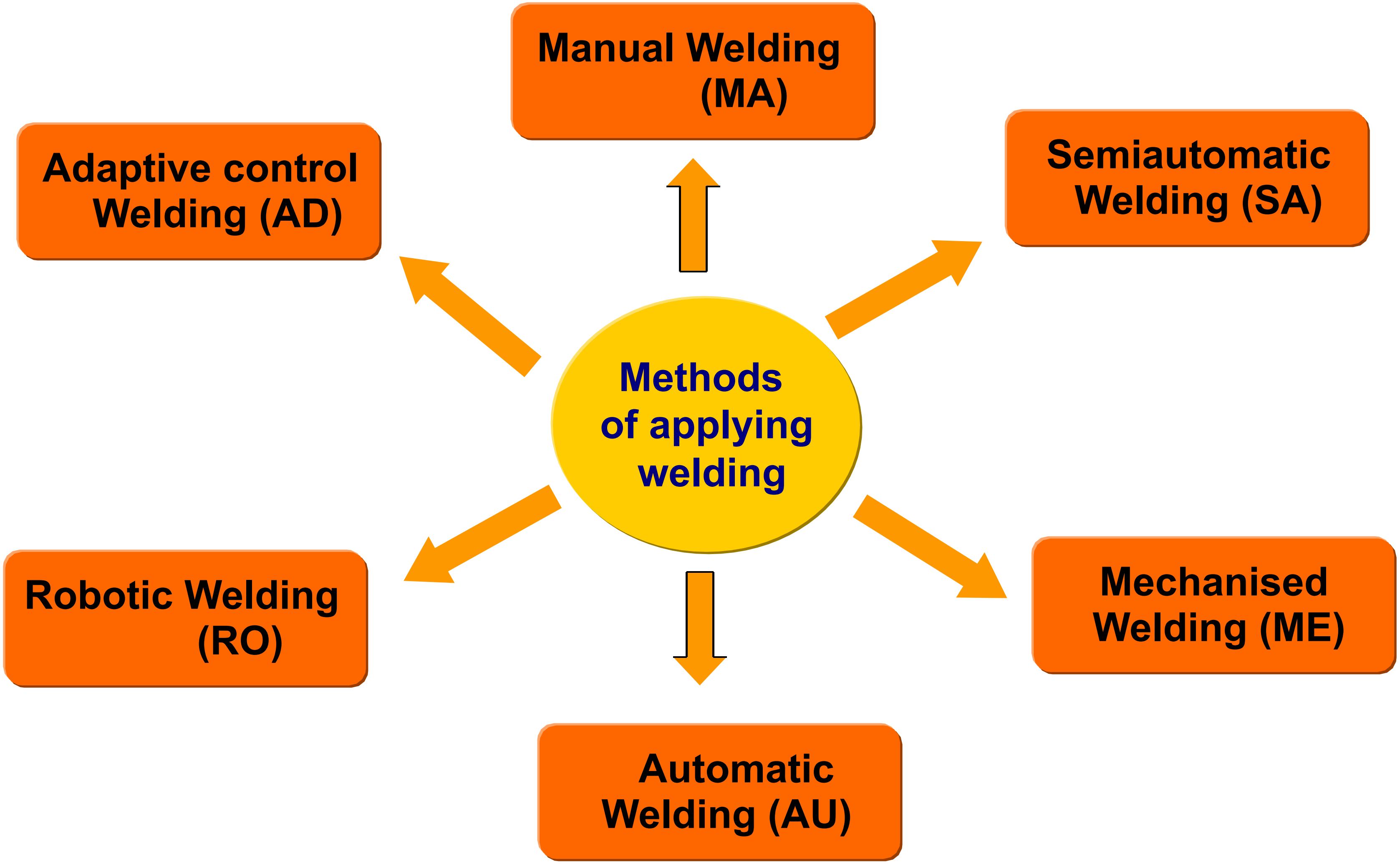
Arc welding → A group of welding processes that produce coalescence of workpieces by heating them with an arc. The processes are used with or without the application of pressure and with or without filler metal.

Oxyfuel welding → A group of welding processes that produces coalescence of workpieces by heating them with an oxyfuel gas flame. The processes are used with or without the application of pressure and with or without filler metal.

Resistance welding → A group of welding processes that produces colescence of the faying surfaces with the heat obtained from resistance of workpieces to the flow of the welding current in a circuit of which the workpieces are a part, an by the application of pressure.

Solid state welding → A group of welding processes that produces colescence by the application of pressure without melting any of the joint components.

Method of applying weld

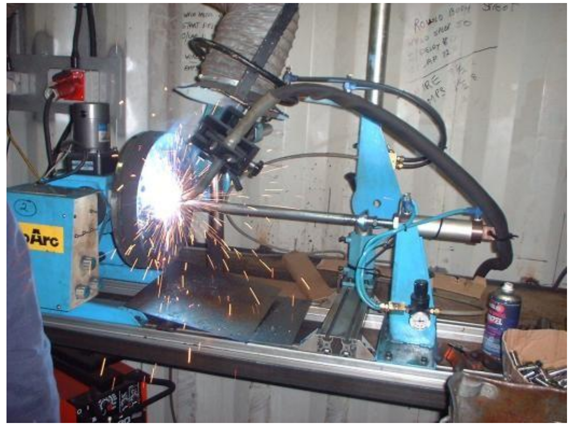


Method of applying weld: Definitions

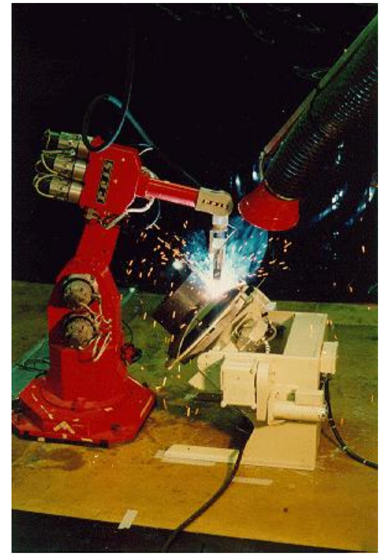
Manual welding (MA)	Welding with the torch, gun, or electrode holder held and manipulated
Semiautomatic welding (SA)	Manual welding with equipment that automatically controls one or more of the welding conditions
Mechanized welding (ME)	Requires manual adjustment of the equipment controls in response to visual observation, with torch, gun or electrode holder by a mechanical device
Automatic welding (AU)	Requires only occasional or no observation of the welding and no manual adjustment of the equipment controls
Robotic Welding (RO)	Welding that is performed and controlled by robotic equipment
Adaptive control welding (AD)	Welding with a process control system that determines changes in welding conditions automatically and directs the equipment to take appropriate action

Method of applying weld: Figures

Manual welding

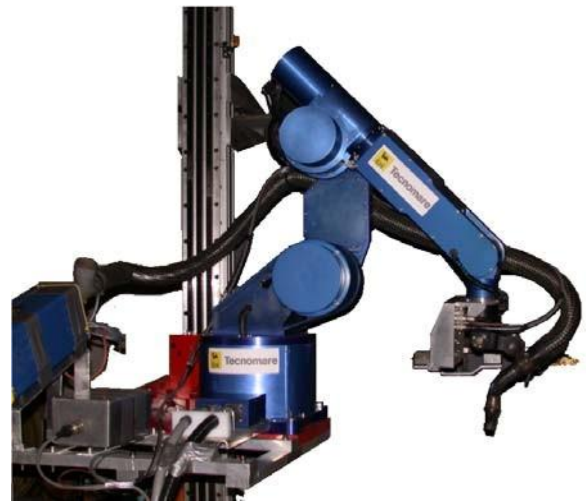


Semiautomatic welding



Adaptive control welding

Automatic welding



Robotic welding

Arc welding with a nonconsumable electrode

- There are two basic types of welding arcs: One uses a nonconsumable electrode and the other uses a consumable electrode.

- The nonconsumable electrode does not melt in the arc, and filler metal is not carried across the arc gap. The welding processes are:
  - Gas tungsten arc welding
  - Plasma arc welding
  - Carbon arc welding

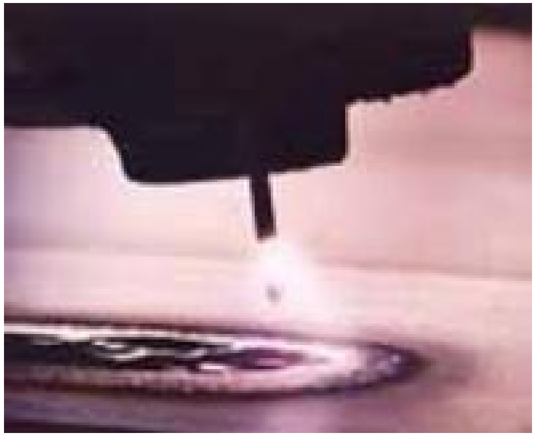
- The main function of the arc is to produce heat. At the same time it produces a bright light, noise, and ionic bombardment that removes the oxide surface of the base metal.

Welding arc

- A welding arc is a sustained high-current, low-voltage electrical discharge through a high conducting plasma that produces sufficient thermal energy which is useful for joining metals by fusion.
- The welding arc occurs between the end of an electrode and a workpiece that carries current.
  - An arc *current* from 1A ---- 3000A   Voltage from 10V ---- 40V

- Whether the electrode is positive or negative, the arc is restricted at the electrode and spread out toward the workpiece.

- The length of the arc gap is proportional to the voltage.
- The arc length for welding is the dimension equal to the electrode diameter, up to about four times the electrode diameter.



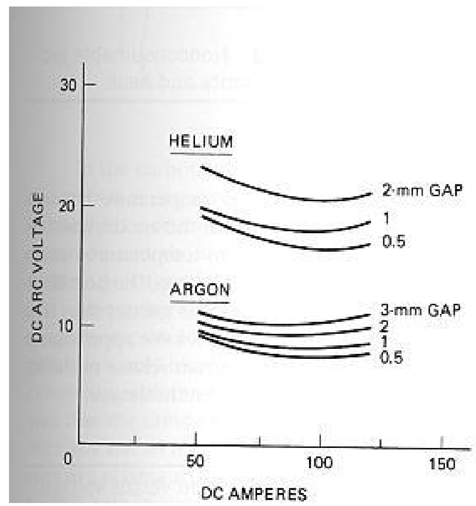
A welding arc

Welding arc

- The arc column is normally round in cross section and is made of two concentric zones: an inner core or plasma and an outer flame.
  - The plasma carries most of the current and has the highest temperature.
  - The outer flame of the arc is much cooler and tends to keep the plasma in the center.

- The temperature and the diameter of the central plasma depend on the amount of current passing through the arc, the shielding atmosphere, electrode size, and type.

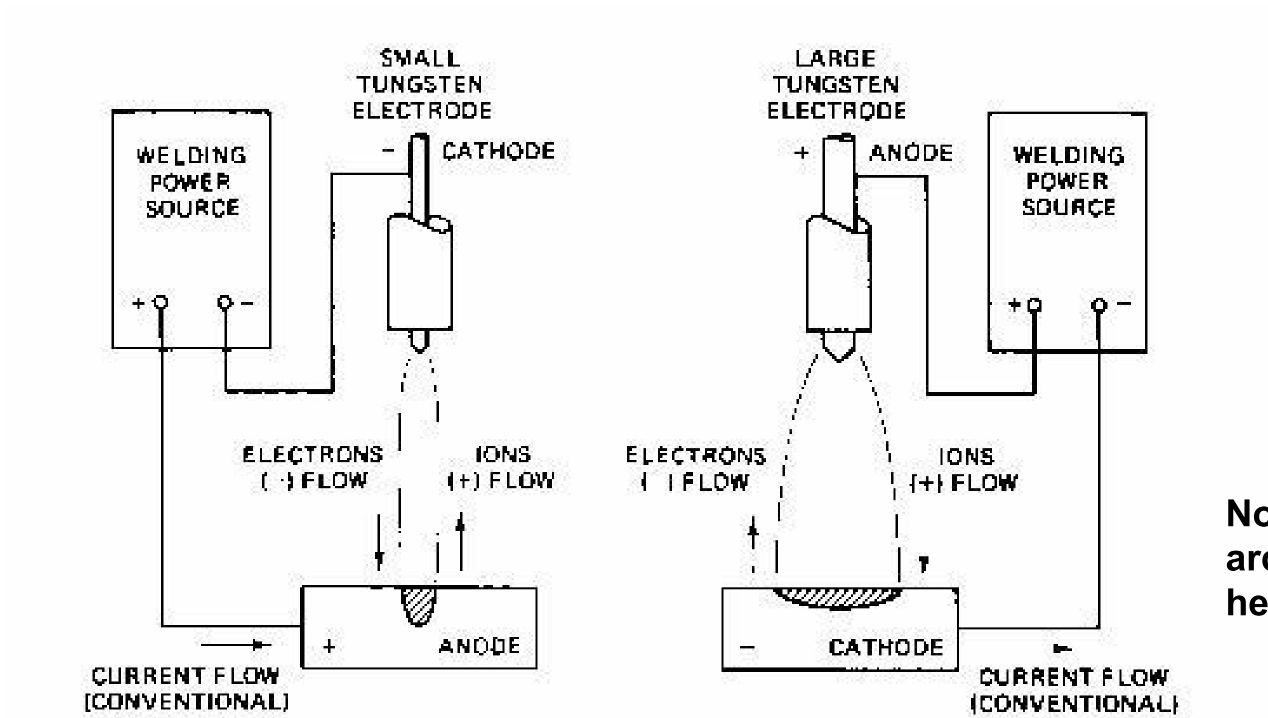
- The relationship between current and arc voltage is not a straight line. In general, the arc voltage increases slightly as the current increases.



Arc characteristics volt-ampere curve

Welding arc

- The arc occurs when electrons are emitted from the surface of the negative pole (cathode) and flow across a region of hot electrically charged plasma to the positive pole (anode), where they are absorbed.



Nonconsumable arc, polarity and heat

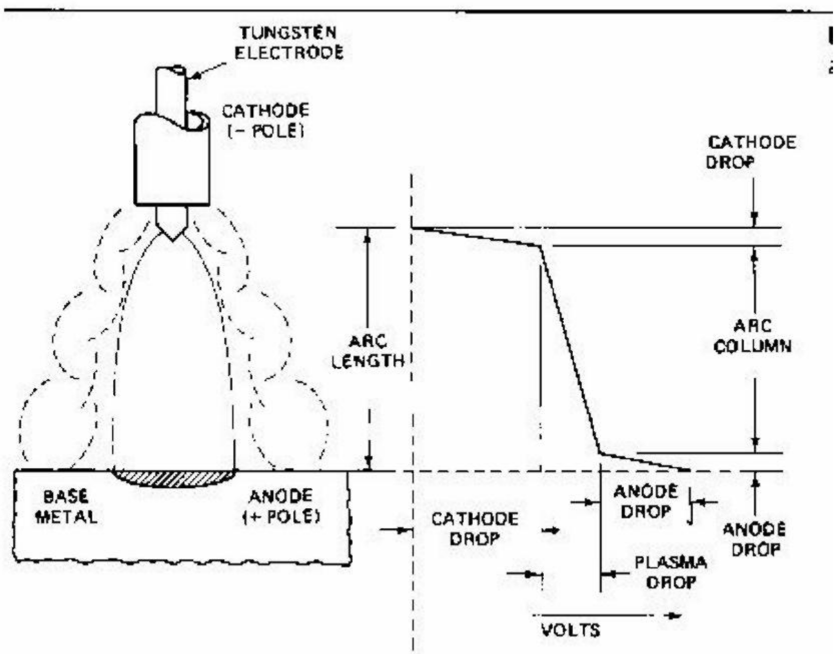
The polarity change affects the weld penetration

DCEN Æ Deep penetration (narrow melted area)

DCEP Æ Shallow penetration (wide melted area)

Welding arc

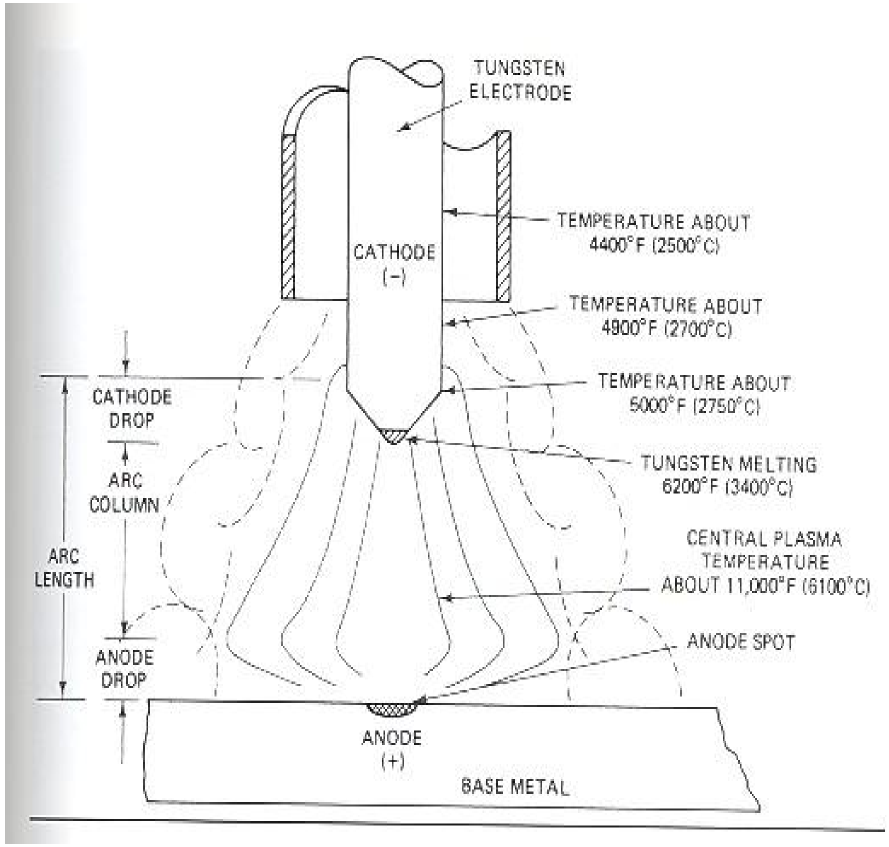
- The arc lenght or gap between the electrode and the work can be divided into three regions: a central region, a region adjacent to the electrode, and a region adjacent to the work.
- At the end regions the cooling effects of the electrode and the work cause a rapid drop in potential, known as anode and cathode drop.
- The length of the central region or arc column represents 99% of the arc length and is linear with respect to arc voltage.



Arc region vs voltage and heat

Welding arc

- The cathode drop is the electrical connection between the arc column and the negative pole (cathode). There is a relatively large temperature and this is the point at which the electrons are emitted through the arc column.
- The stability of the arc depends on the smoothness of the flow of electrons at this point.
- Tungsten and carbon provide thermionic emissions since both are good emitters of electrons.
- They have high melting temperatures, are practically nonconsumable, and are therefore used for welding electrodes. Tungsten has the highest melting point of any metal.
- The anode drop occurs at the other end of the arc and is the electrical connection between the positive pole and the arc column. The temperature changes from the arc column to the anode is considerably lower

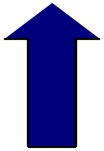
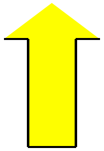
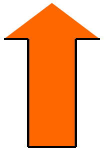


Temperature of a DC tungsten arc

Welding arc: penetration

- The thermal energy generated in the arc is the product of welding current and arc voltage. The heat raises the temperature of the base metal, causing melting and resulting in a molten pool.
- The heat of the arc is distributed through radiation, convection, and conduction to the base metal.

- Penetration of the arc depends on;
  - Polarity of the arc
  - The composition of shielding gas
  - Mass of the base metal and its composition (thermal conductivity and melting temperatures)
  - Preheat temperature

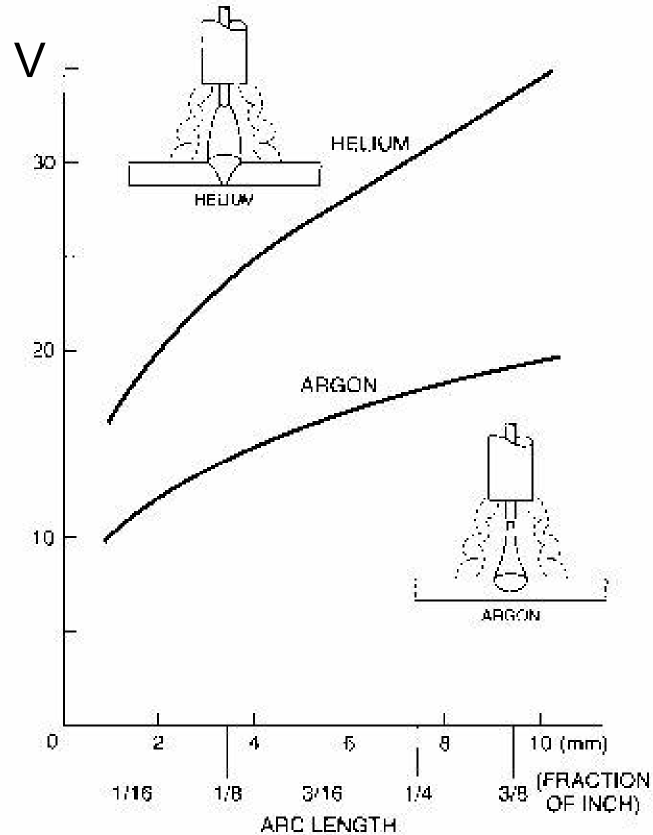
Welding current  Travel speed of current  Depth of penetration 

Welding arc: gas composition

- The gas composition of the area surrounding the arc has an influence on the characteristics of the arc.
- The voltage of a helium-shielded arc is higher than that of an argon-shielded arc for the same length carrying the same current, due to the higher ionization for helium (24,5V).

  Argon ->15,7V.

- The arc shielded with helium has more power (heat) and can do more work. The helium shielded arc column is;
  - **Larger**
  - **More penetration**
  - **Higher travel speed**
  - **Weld heavier base metals**



Arc voltage and arc length

Gas tungsten arc welding (GTAW)

- It is an arc welding process that uses an arc between a tungsten electrode (nonconsumable) and the weld pool.
- The process is used with shielding gas and without the application of pressure.

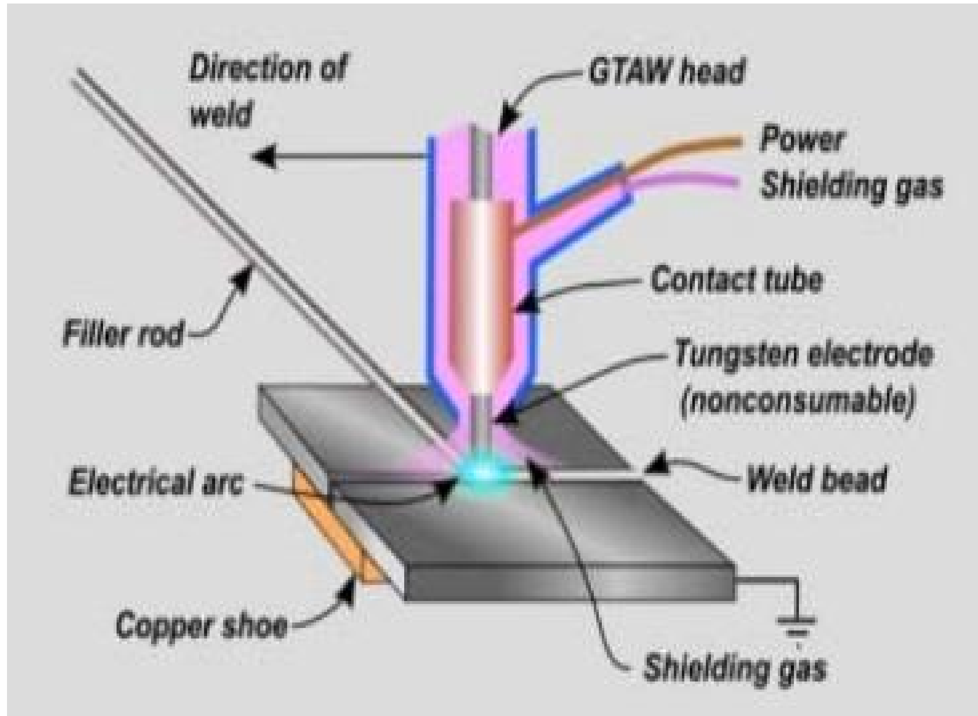
• Filler metal may or may not be used.

• It is also know as;

• **TIG** (Tungsten inert gas) in UK

• **WIG** (wolfram inert gas) in Germany

• **GTAW** in USA



Process diagram for GTAW

GTAW: Principles of operation

- The process utilizes the heat (≈ 6100 ºC) of an arc between a nonconsumable tungsten electrode and the base metal, that is melted to form a melted pool.
- Filler metal is not added when thinner materials, edge joints and flange joints are welded. This is called as autogenous welding.
- For thicker materials an externally fed or cold filler rod is generally used.
- The arc area is protected from the atmosphere by the inert shielding gas flown from the nozzle of the torch.
- The shielding gas displaces the air, so that the oxygen and the nitrogen of the air do not come in contact with the molten metal or the hot tungsten electrode.

- There is little or no spatter and little or no smoke.
- The resulting weld is smooth and uniform and requires minimum finishing.

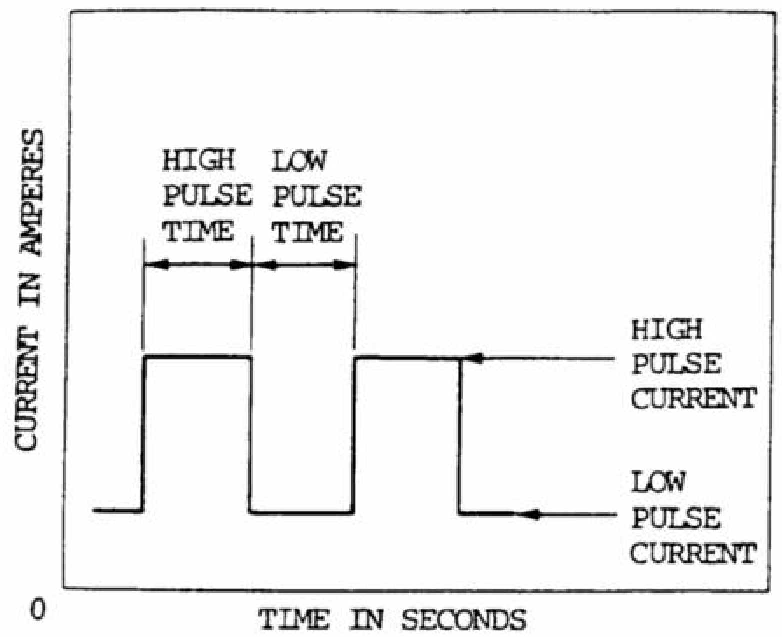
GTAW: Advantages and major uses

- High quality of welds in almost all metals and alloys
- Very little, if any, postweld cleaning is required
- The arc and weld pool are clearly visible to the welder
- There is no filler metal carried across the arc, so little or no spatter
- Performed in all positions
- No slag produced that might be trapped in the weld
- Extreme control for precision work and high quality
- Heat can be controlled very closely and the arc can be accurately directed

- Used for mainly thinner materials
- Very useful for maintenance and repair work
- Welding for unusual metals
- Joining thin wall tubing and making root passes in a pipe joints
- Manual (mostly), mechanised, automatic and semiautomatic (limited) applications are available

GTAW: Variations

- **Pulsed-current GTAW:** the welding current continiously changes between two levels. During the period of high-pulsed current, heating and fusion take place, during the low-pulsed current period, cooling and solidification take place.



Pulsed current welding

- **Manual programmed GTAW:** welding current rise and fall at specific rates to specific values are programmed. A switch mounted on the torch is used to start and stop the program. It is popular for welding tubing and root pass welding of pipe.

Manual programmed

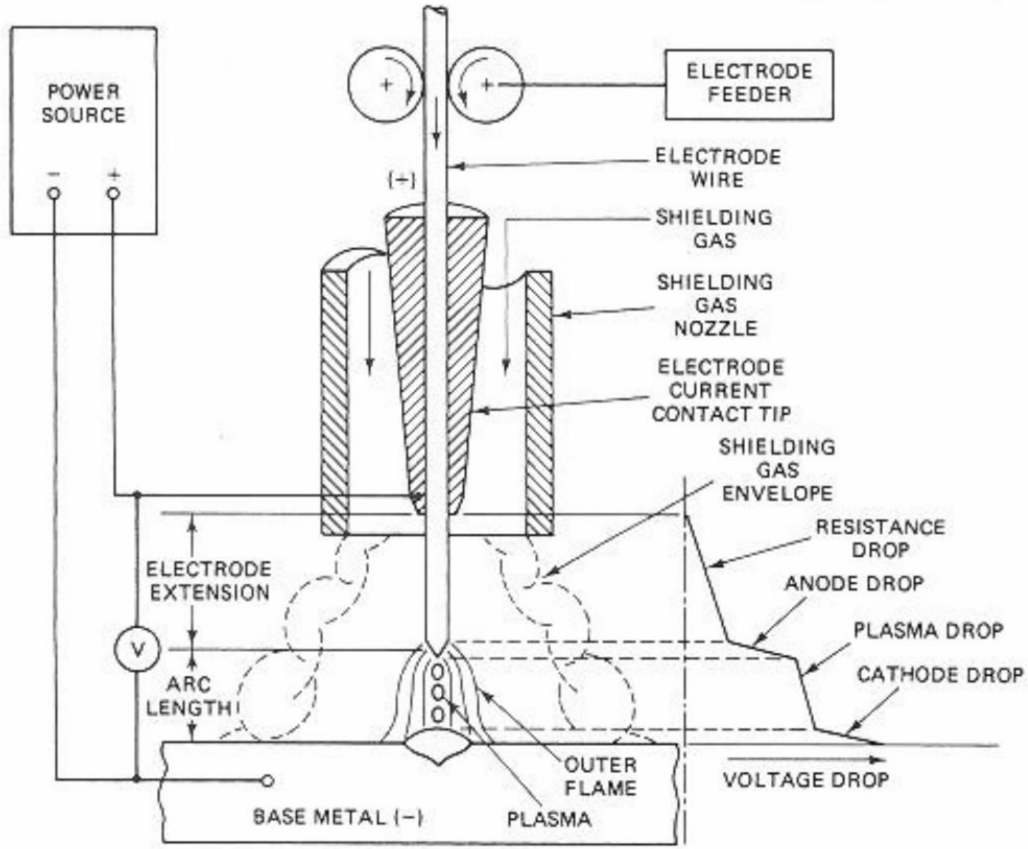
GTAW: Industrial use and typical applications

- The aircraft industry
- Space vehicles fabriction (shells, structures, various tanks)
- Thin-wall tubing
- Root-pass welds in piping
- Repair and maintenance industry
  - Repairing tools and dies
  - Repairing aluminium and magnesium parts
  - Repairing of highly critical items

Arc welding with consumable electrode

- Electrode is melted and the molten metal is carried across the gap.
- A uniform arc length is maintained between the melting end of the electrode and weld pool.
  - Shielded metal arc welding
  - Gas metal arc welding
  - Flux cored arc welding
  - Electro slag welding
  - Submerged arc welding

- The electrode is continiously fed into the arc and is melted by the heat of the arc as a deposition.



Arc region of the consumable electrode arc

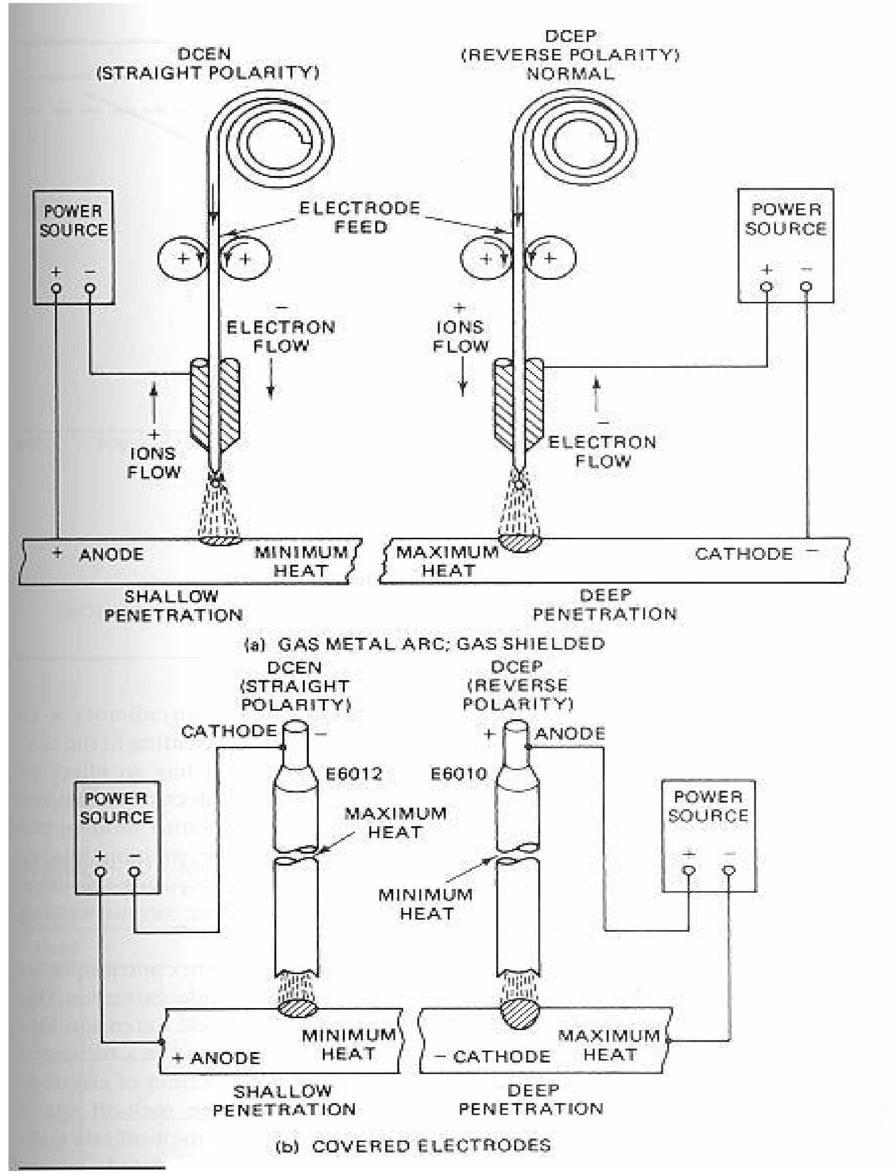
Arc welding with consumable electrode

- Good quality of welding and high-productivity welding depend on two major factors:
  - The penetration of the weld into the base metal
  - The melt-off rate of the electrode
- The maximum heat normally occurs at cathode.
  - When straight polarity welding (DCEN), the melt-off rate is high, but the penetration of the base metal is low.
  - When DCEP welding, the max. heat still occurs at cathode, deep penetration occurs.

High current electrode Æ melt-off rapidly

Low current Æ melt-off slowly

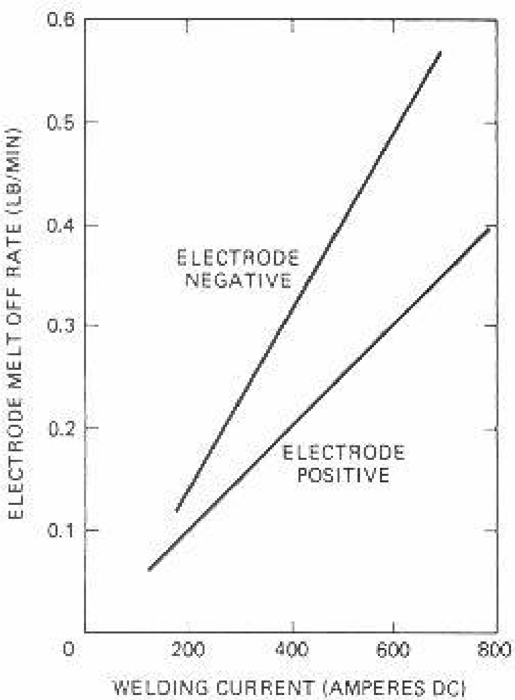
Melt-off: The heat required to melt the electrode is a physical relationship between the current and the weight of metal melted, known as melt-off or burn-off rate which is the weight of metal melted per unir time.

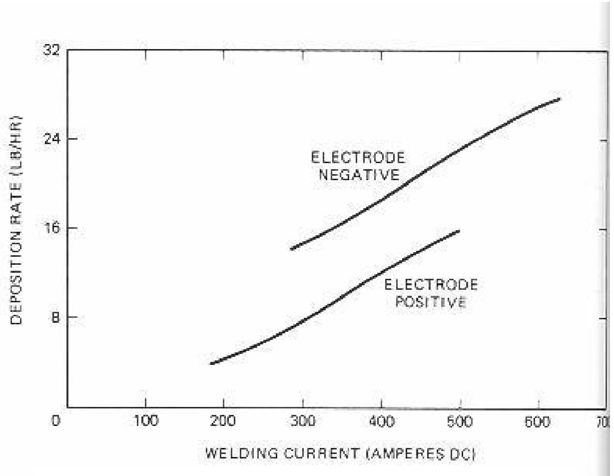


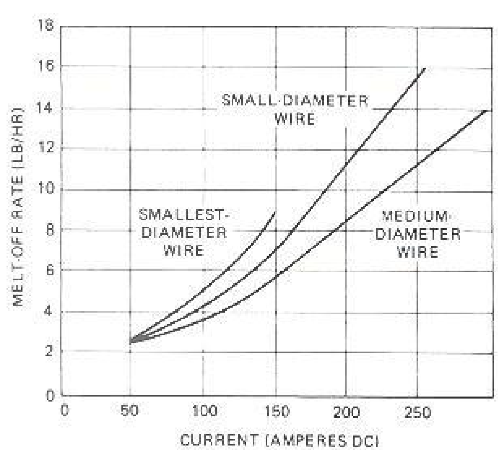
Polarity and heat relationship

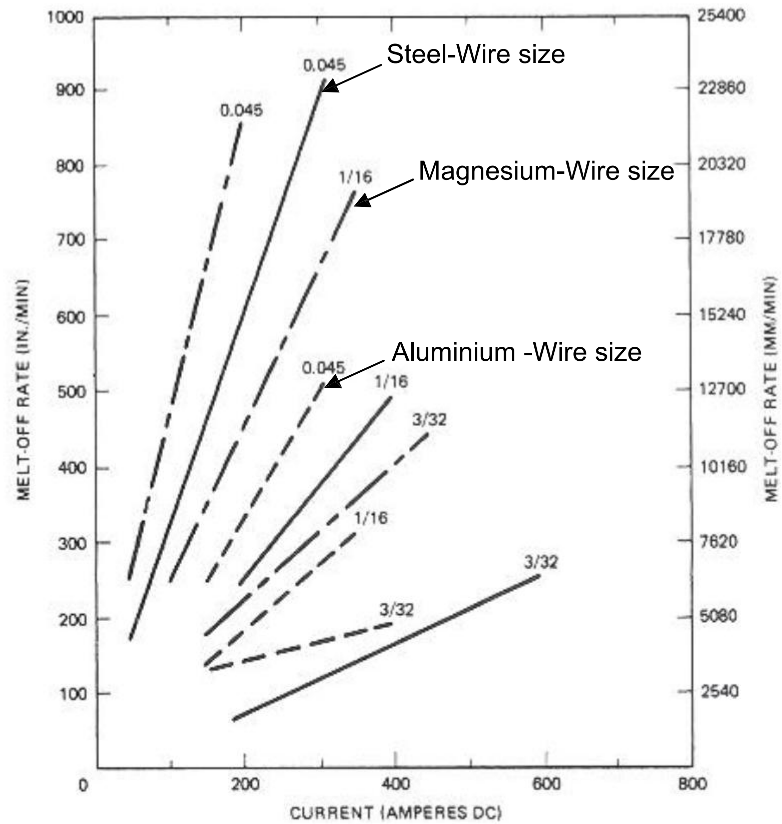
Melt-off rate and its factors

- Factors that affect the melt-off rate:
  - Melting point of the material (i.e. Al has higher melt-of rate)
  - The size of electrode wire: based on current densityÆ I / cross-section area
  - Electrode extension: resistance to heat



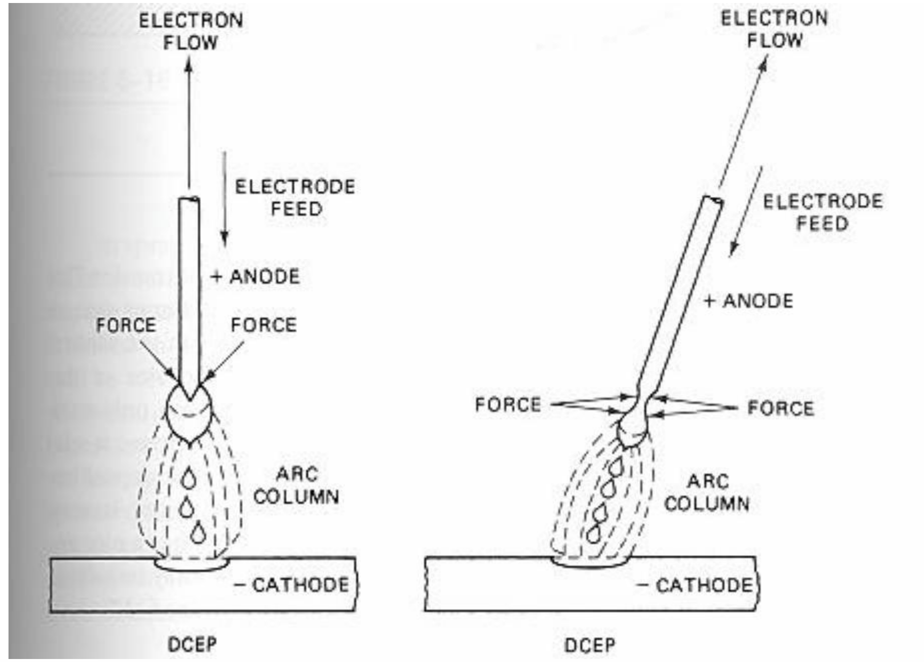






Metal transfer across the arc

- The forces that causes metal to transfer across the arc are similar for all the cosumable electrode arc welding.
- The metal being transferred ranges from small droplets, smaller than the diameter of the electrode, to droplets much larger in diameter than the electrode.
- The mechanism of transferring liquid metal across the arc gap is controlled by:



Electromagnetic force on drop about to transfer

- **Surface tension** → *Causes the surface of the liquid to contract to the smallest possible area*
- **The plasma jet** → *Molten metal drops in flight are accelerated toward the workpiece*
- **Gravity** → *Tends the detach the liquid drop*
- **Electromagnetic force** → *Acts to detach a molten drop at the tip of the electrode.*

•The combination of these forces that acts on the molten droplet and determines the trasfer

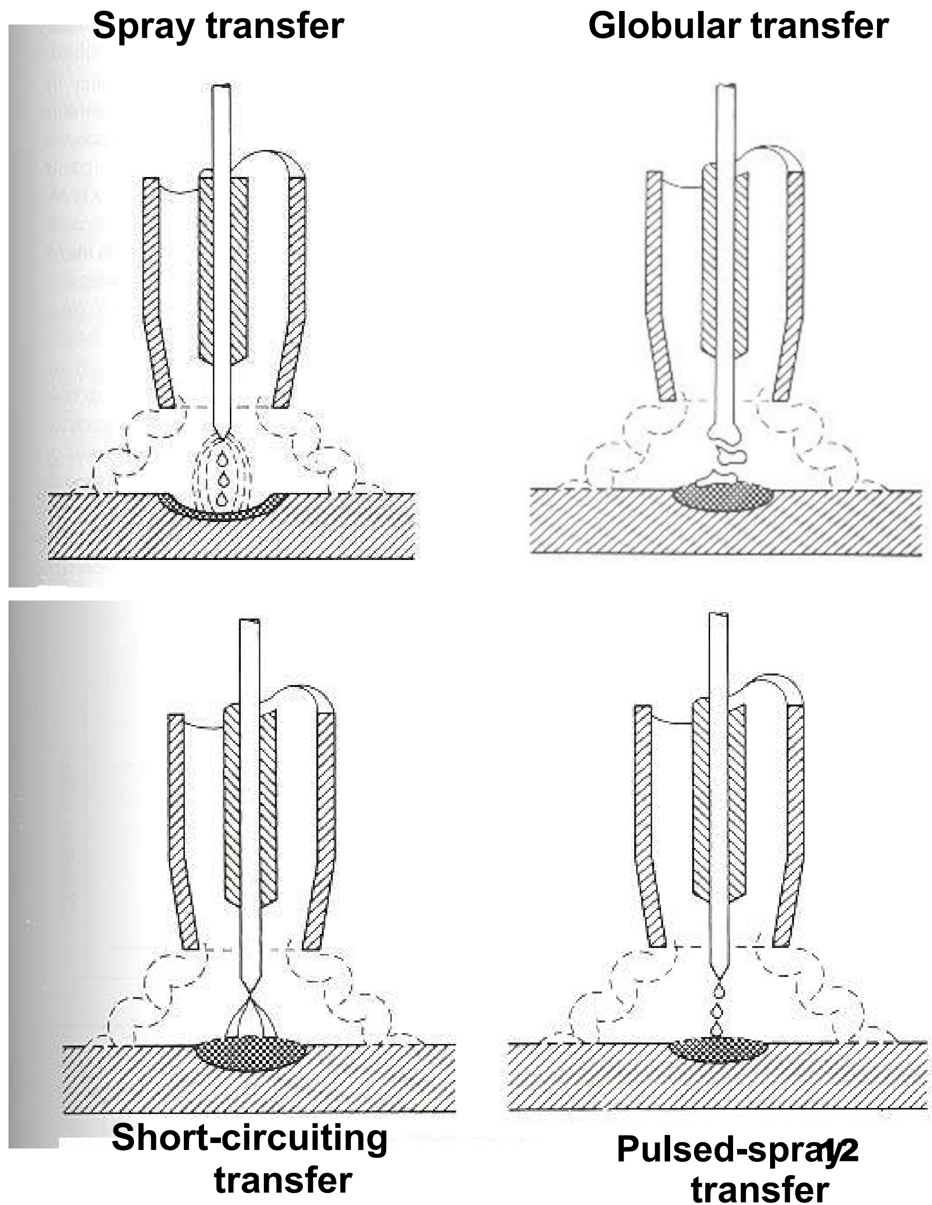
Modes of metal transfer

- The mode of metal transfer across the arc is related to the welding process;
  - The metal involved
  - The arc atmosphere
  - The size, type and polarity of the electrode
  - The characteristics of the power source
  - The welding position
  - Welding current, current density, and heat input
- The most common way to classify metal transfer is according to size and frequency and characteristics of the metal drops being transferred.
- Four major types of metal transfer:
  - Spray transfer
  - Globular transfer
  - Short-circuiting transfer
  - Pulsed-spray metal transfer

There is an intermediate form of transfer in the transition zone between two modes where both types of transfer may occur simultaneously.

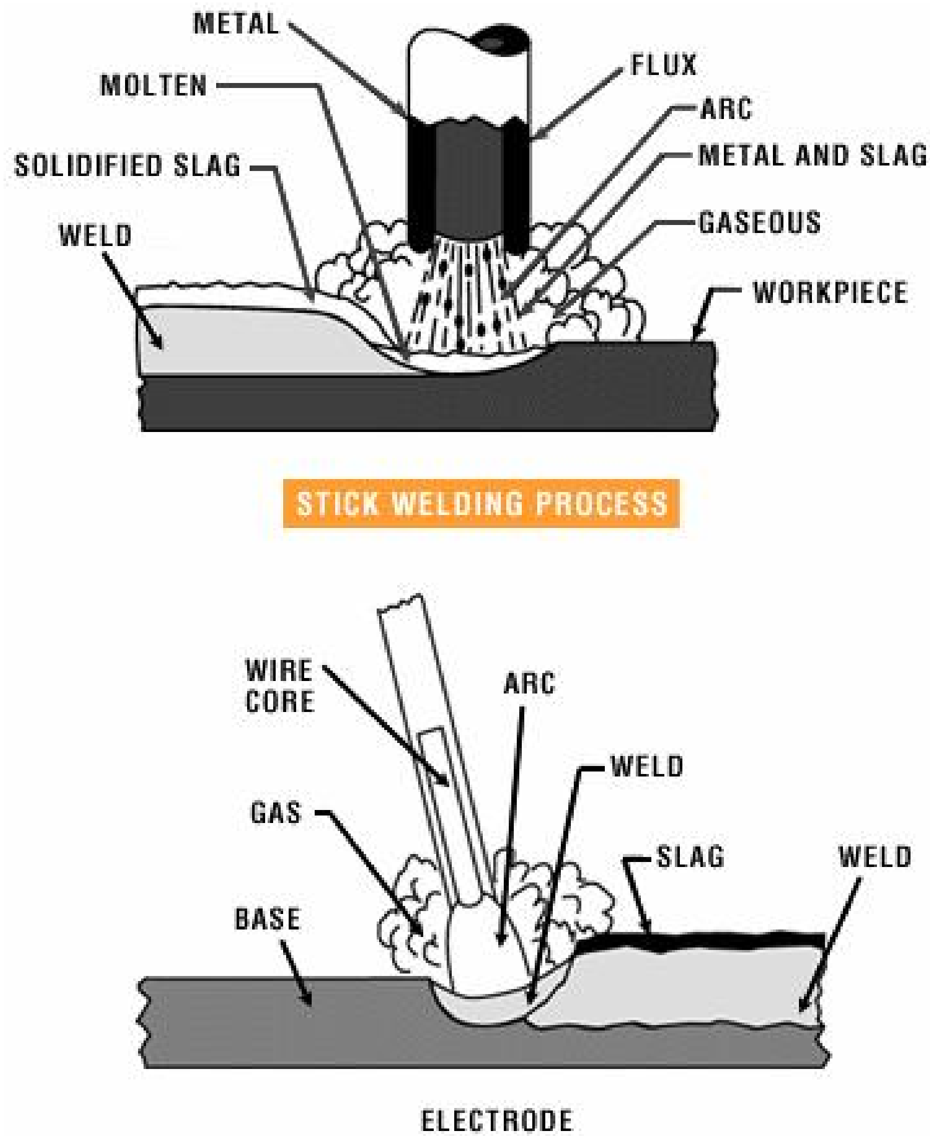
Modes of metal transfer

- **Spray transfer:** The drops of molten metal are approximately the same size as the electrode wire. It occurs in an inert gas atmosphere, %80 argon. *Smooth transfer, large weld pool, good penetration, not used for thin materials*.
- **Globular transfer:** The molten globule can grow in size until its dia reaches 1.5 to 3 x D electrode. It usually occurs when CO_2 shielding gas used. *Very deep penetration, only used in flat pos. Used for heavy steel sections.*
- **Short-circuiting transfer:** It is a low-energy mode of transfer. The molten tip may grow up to 1.5 times the electrode dia. *Weld pool is small, not used on nonferrous metals.*
- **Pulsed-spray metal transfer:** It produces droplets of approximately the same or smaller size than the electrode dia. It is based on a special pulsed waveform of the welding current. Type of pulsing is difficult to adjust, *never become popular.*



Shielded metal arc welding (SMAW)

- SAW is an arc welding process with an arc between a covered electrode and the weld pool.
- The process is used with shielding from the decomposition of the electrode covering, without the application of pressure, and with filler metal from the electrode.
- SAW is also known as *stick electrode welding*.



SMAW: Principles of operation

- It consists of an arc between a covered electrode and the base metal.
- The arc is initiated by touching the electrode momentarily to the workpiece.
- The heat of the arc melts the surface of the base metal to form a molten pool. The metal melted from the electrode is transferred across the arc into the molten pool.
- The size of the weld pool and the depth of penetration determine the mass of molten metal under the control of welder.
- If current is too high, the depth of penetration will be excessive and the volume of molten weld metal will become uncontrollable.
- A higher speed of travel reduces the size of the molten weld pool.
- The weld metal deposit is covered by a slag from the electrode covering.
- The arc in the immediate arc area is enveloped by an atmosphere of protective gas produced by the disintegration of the electrode coating.

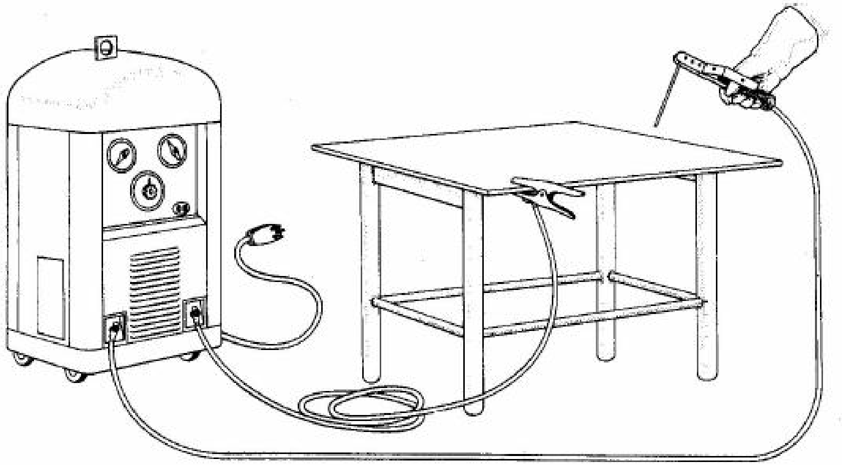
SMAW: Advantages and major uses

- It is the most popular arc welding process.
- It has maximum flexibility and can weld many metals in all positions from near minimum to maximum thickness.
- The investment for equipment is small.
- It is used in manufacturing and in field work for construction and maintenance.
- The method of application is manual. Semiautomatic and mechanised methods are not used. Automatic method can be used.
- Welding in the horizontal, vertical and overhead positions are possible depends on the type and size of the electrode, as well as the welding current and the skill of the welder.

SMAW: Equipment required

- The welding machine or power source: to provide electric power of the proper current (CC, 25A-500A) and voltage (15 to 35V).

- Electrode holder, held by the welder. It firmly grips the electrode and transmits the welding current to it.

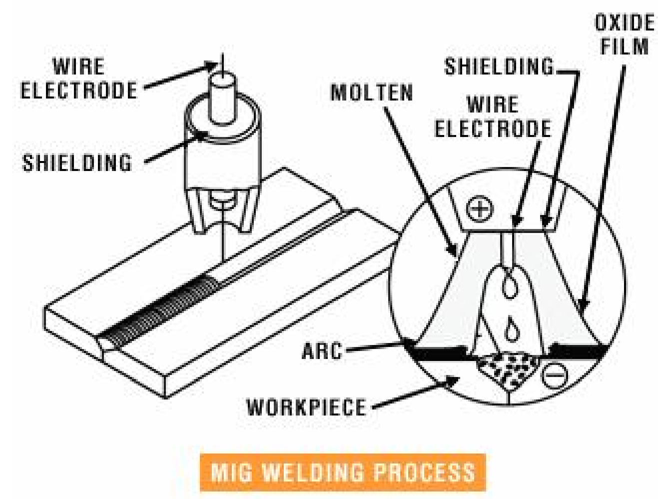


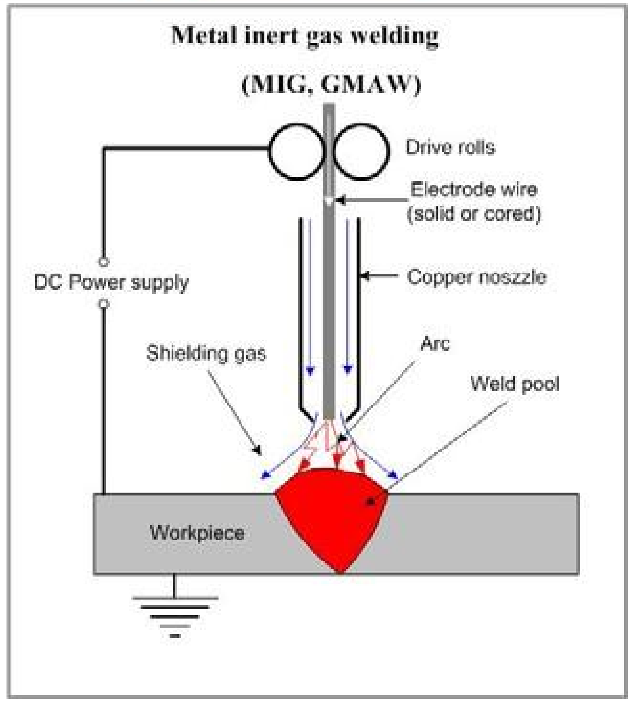
SMAW: Material used

- The covered electrode is the only item of material normally required.
- The selection of the covered electrode is based on the electrode usability and the composition and the properties of the deposited weld.
- The coating on the electrode provides:
  - gas from the decomposition of certain ingredients of the coating to shield the arc from the atmosphere.
  - the deoxidizers for scavenging and purifying the weld deposited
  - slag formers to protect the deposited weld
  - ionzing elements to make the arc more stable
  - alloying elements to provide special characteristics to the deposited weld
  - iron powder to improve productivity of the electrode.

Gas metal arc welding (GMAW)

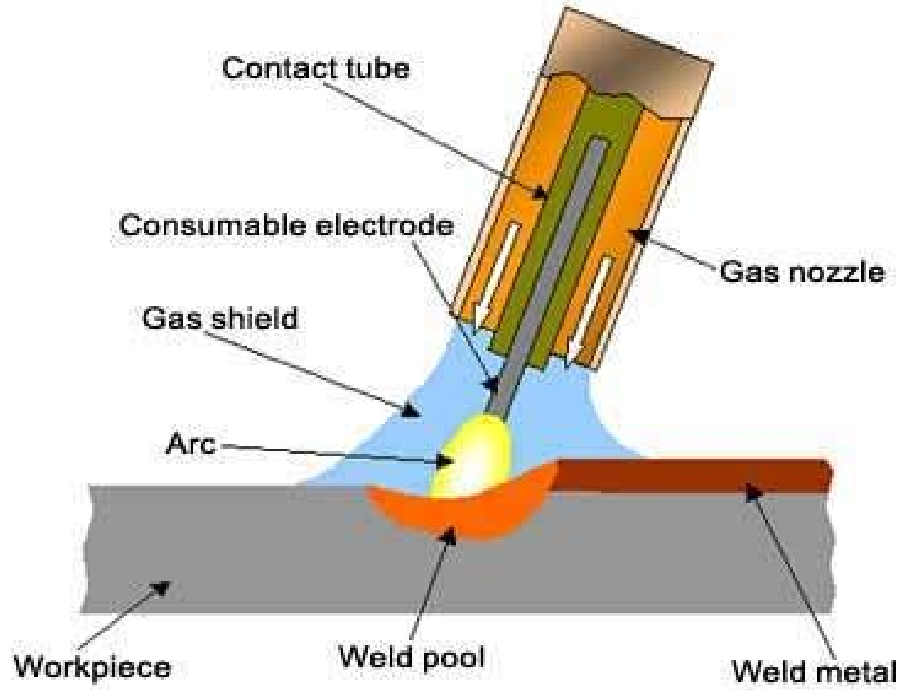
- It is an arc welding process that uses an arc between a continious filler metal electrode and the weld pool.
- The process is used with shielding from an externally supplied gas and without the application of pressure.
- This is also called as metal inert gas (MIG) or metal active gas (MAG) welding.
- There are many variations depending on the type of shielding gas, the type of metal transfer, the type of metal welded etc.





MIG: Principles of operation

- MIG welding utilizes the heat of an arc between a continiously fed consumable electrode and the work to be welded.
- The heat of the arc melts the surface of the base metal and the end of the electrode.
- The metal melted off the electrode is transferred across the arc to the molten pool.
- The penetration is mainly controlled by welding current.
- The width of the molten pool is mainly controlled by the travel speed.
- Shielding of the molten pool, the arc, and the surrounding area is provided by an envelope of gas fed through the nozzle.



- The shielding gas may be an inert gas, an active gas, or a mixture, surrounds the arc area to protect it from contamination from the atmosphere.

MIG: Advantages and major uses

- MIG welding is one of the most popular arc welding process.
- Continuous wire feed
- Automatic self-regulation of the arc length
- High deposition rate and minimal number of stop/start locations
- Welder has good visibility of weld pool and joint line
- Little or no post weld cleaning
- Can be used in all positions
- Wide range of application: sheet metal industry, pipe welding.

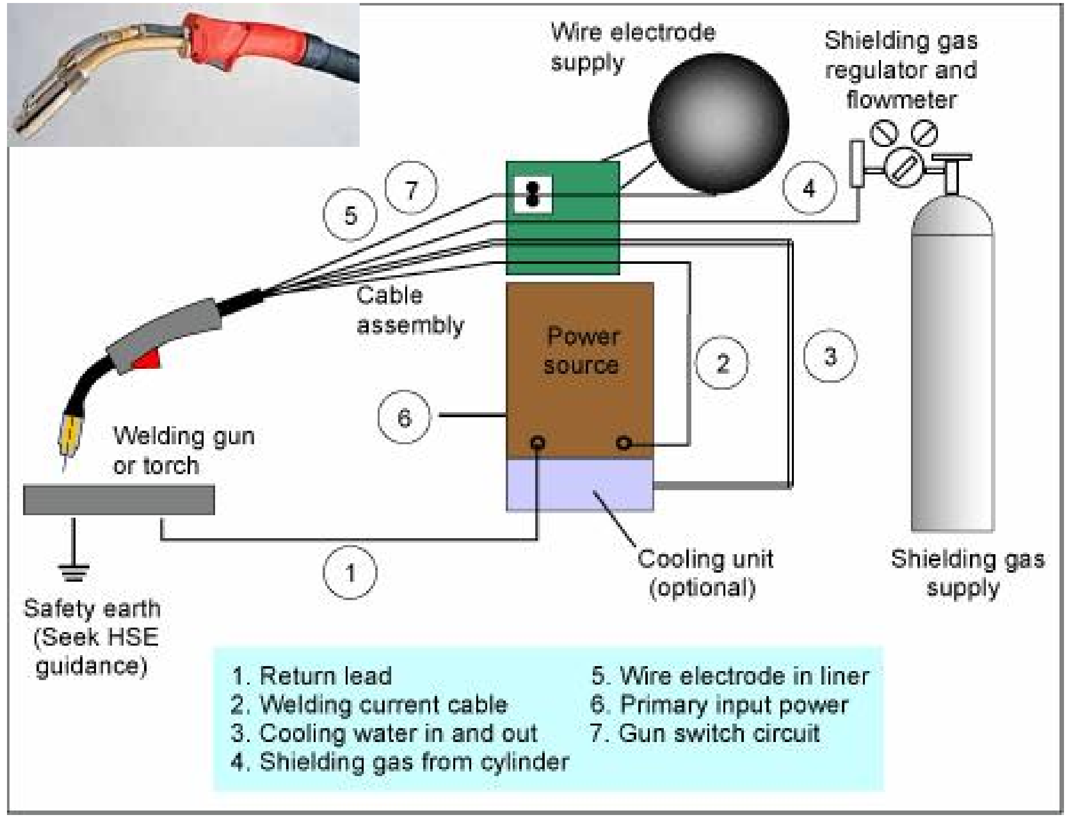
Disadvantages:

- High level of equipment maintenance
- No independent control of filler addition
- Lower heat input can lead to high hardness values
- Joint and part access is not as good as TIG (tungsten inert gas) welding

- The MIG process uses semiautomatic, mechanised, or automatic equipment.
- In semiautomatic welding, the wire feed rate and arc length are controlled automatically.
- In mechanised welding, all parameters are under automatic control.
- With automatic equipment, there is no manual intervention during welding.

MIG: Equipment required

- A MIG system consists of:
  - Power source, 50-500A and 10-50V
  - The electrode wire feeder and control system
  - The welding gun and cable assembly for semiautomatic welding or the welding torch for automatic welding
  - The gas and water control system for the shielding gas and cooling water
  - Travel mechanism and guidance for automatic welding



- MIG welding requires high current at a relatively low voltage.
- Three generic types of power source are suitable for MIG welding: AC/DC transformer rectifier or inverter, and DC generator.
- Wire feed unit is to feed the consumable wire at a constant rate.
- High performance feed units are capable of delivering wire at up to 30m/min.